

Forward Models for Nuclear Magnetic Resonance in Carbonate Rocks

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ABSTRACT

In the conventional approach to the interpretation of nuclear magnetic resonance (NMR) measurements on water-saturated reservoir rocks, it is assumed that the T_2 distribution and the pore size distribution are directly related. However, both laboratory and log data show that this relationship breaks down in many pore systems, especially carbonates, which consist of micro (intragranular) and macro (intergranular) porosity. This breakdown limits our ability to predict permeability and movable fluid fractions. Physically, it is due to the diffusion of magnetization between the intra- and intergranular pores.

We present here three geometrical models that help clarify the relationship between NMR measurements and the underlying pore geometry. All of the models characterize this geometry in terms of four parameters: (1) the volume fraction of total porosity, ϕ

- (2) the volume fraction of intergranular porosity, f_m
- (3) the pore volume to surface area ratio for the micropores, $V_{S\mu}$
- (4) the pore volume to surface area ratio for the macropores, V_{Sm}

In the first model, we apply random walk numerical simulations to an ordered cubic packing of consolidated microporous grains. For given values of the above parameters, the T_2 distribution is evaluated as a function of surface relaxation parameter, ρ . In the second and third models, the microporous grains are treated as a continuum. For the ρ values of greatest interest, roughly, $1.50 \rightarrow 7.50 \mu\text{m/s}$, essentially identical results can be derived from a three-dimensional (3D) analytical model. In addition, for all values of ρ , many features of the T_2 distribution can be represented in terms of a one-dimensional (1D) model pore space.

INTRODUCTION

Conventional interpretation of NMR relaxation in water-saturated reservoir rocks is based on two assumptions. The first is that the magnetization relaxation within each pore is controlled by surface relaxivity so that the effective lifetime is V_S / ρ . Here V_S is the volume to surface area ratio of the individual pore and ρ is the surface relaxivity parameter. The net transverse relaxation rate for that pore is then

$$\frac{1}{T_2} = \frac{\rho}{V_S} + \frac{1}{T_{2B}}, \quad (1)$$

where the second term on the right side represents a small correction associated with the bulk lifetime, T_{2B} , of the fluid. Second, in the rock as a whole, each pore is assumed to act independently so that the measured magnetization is simply a sum of contributions from the individual pores (Kenyon et al., 1989, Kenyon, 1992, Kenyon, 1997). In this scenario, it follows that the T_2 and pore size distributions are directly related. In carbonates, especially in grainstones and packstones, experiments performed at our laboratory indicate that these assumptions often fail. The purpose of this paper is to show that this failure can be understood in terms of the diffusion of water molecules between the inter- and intragranular pores. To illustrate, we

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consider in Figures 1 and 2 two examples of Middle East carbonates. In the left-hand panel of Figure 1 we have an image of a peloidal grainstone obtained from an optical thin section. The right-hand panel shows an environmental scanning electron microscope (ESEM) image of the interior of what appear as dark solid grains in the left-hand optical image. Clearly, the large grains are comprised of roughly micron sized pellets and the rock's porosity is divided between two populations with very different length

scales. Similar behaviour is seen in Figure 2 where two ESEM images of a different Middle Eastern carbonate are shown (Anselmetti et al., 1998). Here the left-hand panel shows both large textured grains and intergranular porosity while the right-hand panel illustrates the fine grained character found in the large grains. Despite the fact that these pore spaces involve widely separated length scales, we see in Figure 3 that the T_2 spectrum measured on the sample shown in Figure 1 has only a single peak. Clearly, in this

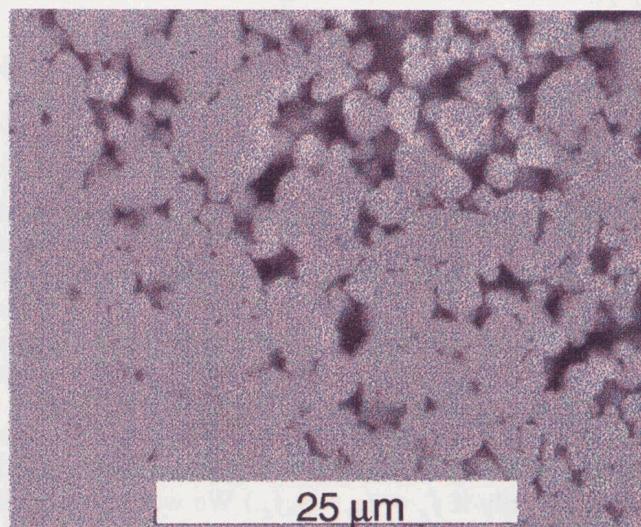
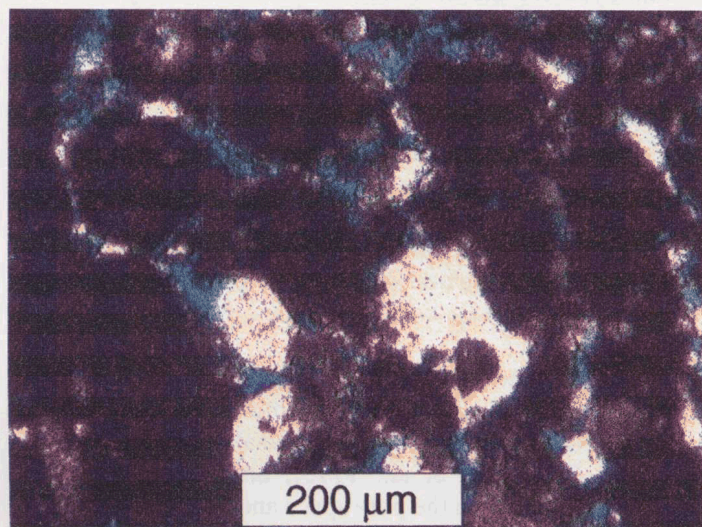


FIG. 1. The left-hand panel is an optical micrograph of a peloidal grainstone with grains that are roughly 200 μm in diameter. Here the white objects are recrystallized calcite grains, the dark objects are microporous grains and the intergranular pores are filled with blue epoxy. The right-hand panel is an environmental scanning electron microscope (ESEM) image taken from the same sample. The fine structure within what appear to be dark solid grains (on the left) is typical of many carbonate grainstones.

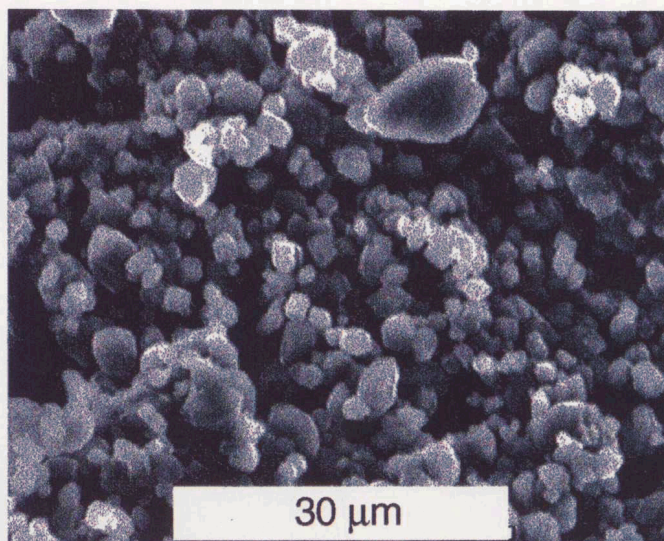
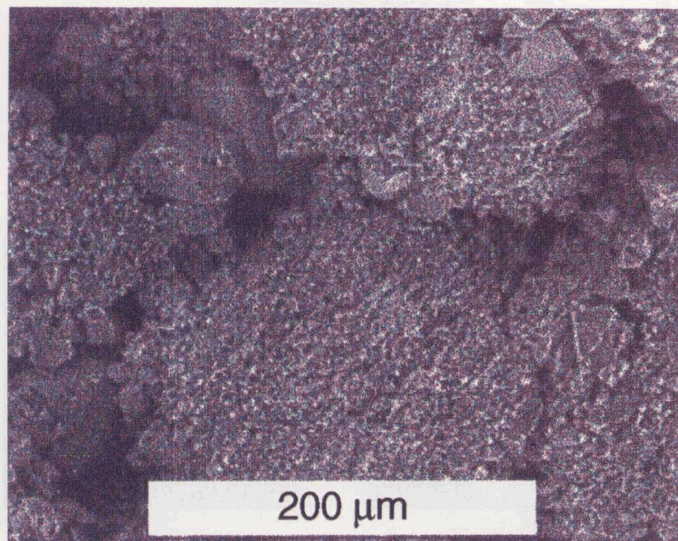


FIG. 2. The left-hand panel is an ESEM image of a peloidal grainstone with a 200 μm scale bar at its bottom. The right-hand ESEM image is from a mudstone sample and the scale bar now represents 30 μm . The fine structure in the mudstone is quite similar to the features within the microporous calcite grains shown on the left.

case the direct link between T_2 and pore size is lost. This may be the reason that carbonate T_2 spectra have been unreliable for estimating both permeability and bound fluid volume.

In the following sections we will develop three models that lead to a common understanding of NMR measurements in water-saturated carbonates with inter- and intra-granular porosity. From the outset we picture the homogeneous rock fabric as having three subvolumes: grains, intergranular pores and vugs, with volume fractions f_g , f_m and f_v , respectively. The grains themselves have porosity ϕ_μ , so the total porosity, ϕ , is given by

$$\phi = f_v + f_m + \phi_\mu f_g, \quad (2)$$

where

$$1 = f_v + f_m + f_g. \quad (3)$$

Vugs are associated with grains that have been dissolved away from the original granular network. Their size can vary from that of a single grain (roughly, 100 μm) to centimeters. Water in vugs contributes to the rock's NMR signal at the bulk fluid rate. Their influence will not be treated explicitly here; if the vug porosity is known a priori, a correction to our *forward* models is easily made. (By contrast, if we are concerned with the inverse problem, the vug contribution to the magnetization decay spectrum can be readily identified only if $f_g \ll f_m + \phi_\mu f_g$.) We will also discuss, briefly, the effects of oil in the pore space, within the framework of a rock that remains water wet.

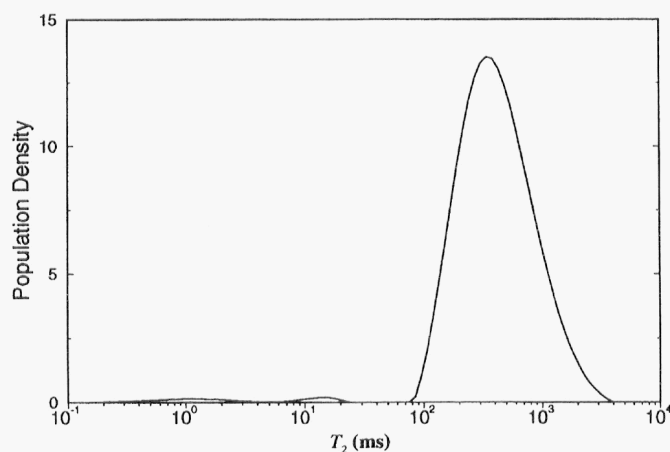


FIG. 3. T_2 distribution measured on a plug taken from a sample similar to the one shown in Figure 1. For this sample the NMR and buoyancy porosities are 24.7 and 25.4 p.u., respectively.

THREE-DIMENSIONAL NUMERICAL SIMULATIONS

The purpose of numerical simulations is to provide results that mimic as closely as possible the physical processes underlying NMR measurements on rocks. They are valuable because (1) they help clarify the basic mechanisms operating within a specified pore geometry and (2) they provide the benchmark for the analytical models to be discussed in the following sections. The analytical models are computationally much less demanding, and can thus be employed for the inversion of both laboratory and field data. The geometry for the numerical simulations (Figure 4) is based on a simple cubic grain consolidation model (Schwartz, 1990) in which each of the large grains comprises smaller, consolidated, grains whose centers lie on a commensurate sublattice. This model is defined by four geometrical parameters: the radii R_g and r_g of the large and the small spheres (which determine the porosity contribution from inter- and intragranular pores) and the lengths, A_g and a_g , of the large and small unit cells. Within the normal range of R_g and r_g , both pore types will form connected pathways.

To study the NMR properties of this geometric model, we have carried out numerical simulations based on random walk techniques. In conventional random walk simulations (Bergman, et al., 1995), diffusing particles are randomly placed in the pore space and take steps of a fixed size, ϵ . With each step, the clock advances by a fixed time, $\epsilon^2/(6D_0)$, where D_0 is the diffusion coefficient for bulk wa-

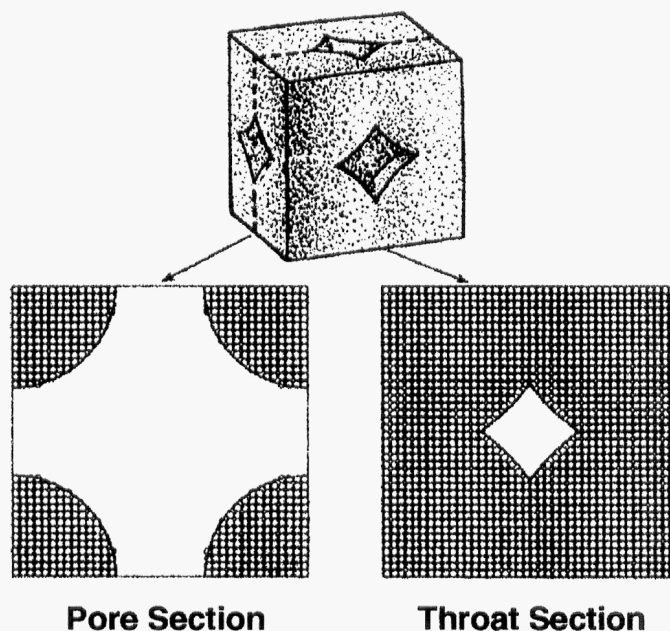


FIG. 4. Three-dimensional model based on a periodic packing of consolidated microporous grains.

ter. In the present case, however, such simulations are not practical. Because ε must be chosen to be much smaller than the smallest features in the pore space (i.e., the throats in the microporous grains) the description of diffusion in the large intergranular pores is prohibitively inefficient.

Accordingly, we have adapted the *first arrival* techniques employed by Torquato and Kim (1989). In this method, at each step one draws the largest possible sphere within the pore space centered at the walker's current position. The walker then jumps to a randomly chosen point on the sphere's surface and the clock is advanced by an amount related to the jump distance. Since we are dealing with diffusion in free space, the distribution of allowed times for a first arrival to the surface of a sphere of radius R can be obtained analytically. In terms of the dimensionless variable $x = D_0 t / R^2$, suppose we denote this distribution as

$f(x)$. ($f(x)$ will be peaked at roughly $x = 1/6$.) It can then be shown that

$$\begin{aligned} P(x) &= \int_0^x f(x_1) dx_1 \\ &= \sqrt{\frac{4}{\pi x}} \sum_{n=0}^{\infty} \exp\left[\frac{-(1+2n)^2}{4x}\right] \\ &= 1 + 2 \sum_{m=1}^{\infty} (-1)^m \exp[-(\pi m)^2 x] \end{aligned} \quad (4)$$

where the series expansions in the second and third lines of this equation are to be used for small and large values of x , respectively. This process continues until the walker reaches a thin layer of width, say, 3ε from the pore grain interface, at which point we revert to the standard random walk algorithm with fixed steps of length ε . If one of these steps takes the walker into the grain, the walk ends with probability γ . When the walk ends, the net time associated with the random walk is entered into a histogram from which one derives the final magnetization versus time curve. The relation between γ and the macroscopic relaxivity parameter ρ is $\rho = 3\gamma D_0 / (2\varepsilon)$ (Bergman et al., 1995).

T_2 distributions obtained from our simulations are illustrated in Figure 5. Here we have taken $A_g = 80 \mu\text{m}$, $a_g = 2 \mu\text{m}$, and $2R_g/A_g = 1.23 = 2r_g/a_g$. These values of R_g and r_g lead to a total porosity, $\phi = 0.31$, which is divided between microporosity, $\phi_\mu f_g = 0.17$, and intergranular porosity, $f_m = 0.14$. In the right-hand panel, the computed T_2 distributions are shown for five values of ρ ranging between $1.50 \mu\text{m/s}$ and $937.5 \mu\text{m/s}$. In the left-hand panel the corresponding results are shown for simulations in which diffusion between the intra- and intergranular pore is forbidden.

It should be emphasized that only the two smallest values of ρ employed in Figure 5 are thought to be representative of reservoir carbonates. Results with larger ρ values are shown to illustrate the physical processes at work. The horizontal axis is ρT_2 which can be thought of as an effective pore size. In both panels, beginning with the shortest times, we have shaded the porosity fraction associated with the microporosity. Note that on the left-hand side the two pore populations remain essentially unchanged as the value of ρ decreases so that the T_2 distribution does, in fact, give an accurate representation of the pore size distribution. By contrast, on the right-hand side, the results are quite different and, as the values of ρ decrease toward the range of experimental interest, two important trends are evident.

First, the area under the short time peak gradually decreases and second, the position of the upper peak shifts toward shorter times. Acting together, these two effects tend to merge the two peaks and produce a T_2 distribution that bears little resemblance to the original pore size distribu-

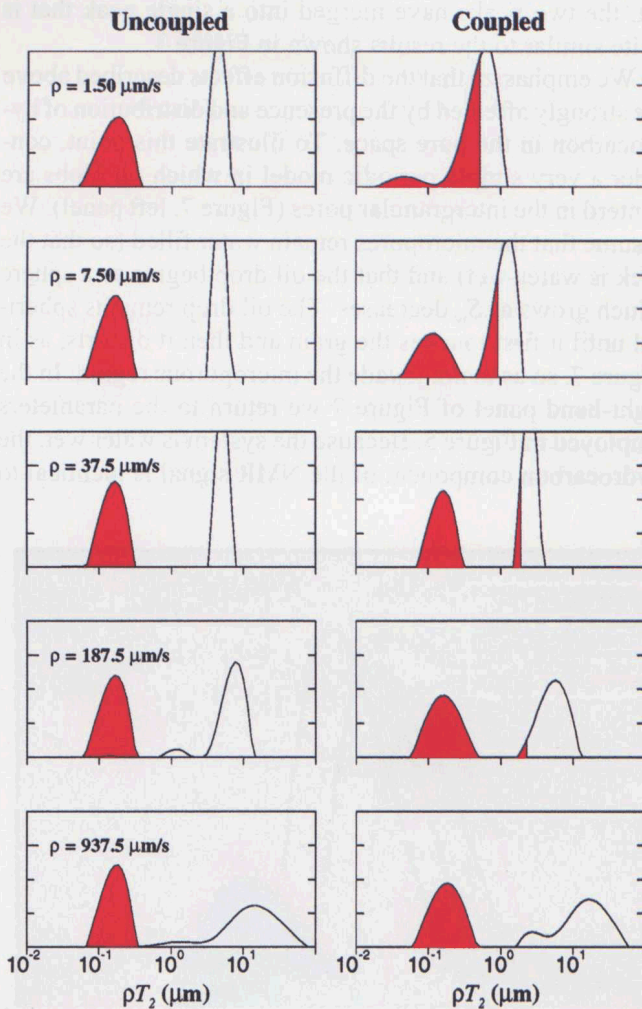


FIG. 5. T_2 distributions are shown for five values of the relaxation parameter, ρ . In the left panel the inter- and intragranular porosity are isolated while in the right panel diffusion between the two parts of the pore space is allowed.

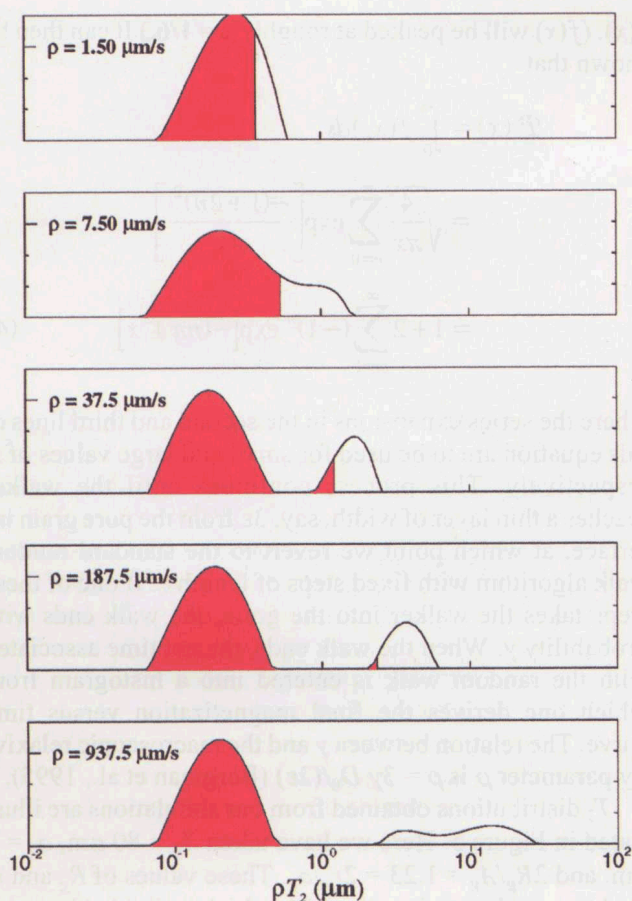


FIG. 6. T_2 distributions are shown (for the same five ρ values used in Figure 5) for a model with $\phi = 0.25$, $f_m = 0.05$, and $\phi_\mu f_g = 0.20$.

tion. Physically, the origin of these effects is diffusion of water molecules between the micro- and macropores. If the value of ρ is small enough, molecules originally in the micropores can diffuse into the macropores before their nuclear spins relax. The decay of these spins then proceeds much more slowly and they contribute to the late time peak. This explains the apparent *transfer* of weight from the early time to the late time peaks. Similarly, molecules originally in the large pores can diffuse into the micropores where they are exposed to an enormous surface compared to the case of a solid grain. This excess surface area leads to increased relaxation rates and thus shifts the position of the late time peak to earlier times.

In Figure 6 results are shown for a system with the same values of A_g and a_g but with $R_g/A_g = 1.30$ and $r_g/a_g = 1.20$ to give $\phi = 0.25$, $\phi_\mu f_g = 0.20$, and $f_m = 0.05$. These parameters were typical of many Middle East carbonates examined in our laboratory. Here, for the ρ values of experimental interest, the two peaks have merged into a single peak that is quite similar to the results shown in Figure 3.

We emphasize that the diffusion effects described above are strongly affected by the presence and distribution of hydrocarbon in the pore space. To illustrate this point, consider a very simple periodic model in which oil blobs are centered in the intergranular pores (Figure 7, left panel). We assume that the micropores remain water filled (so that the rock is water-wet) and that the oil drop begins as a sphere which grows as S_w decreases. The oil drop remains spherical until it first touches the grain and then it distorts, as in Figure 7, so as to not invade the microporous region. In the right-hand panel of Figure 7 we return to the parameters employed in Figure 5. Because the system is water wet, the hydrocarbon component of the NMR signal is identical to

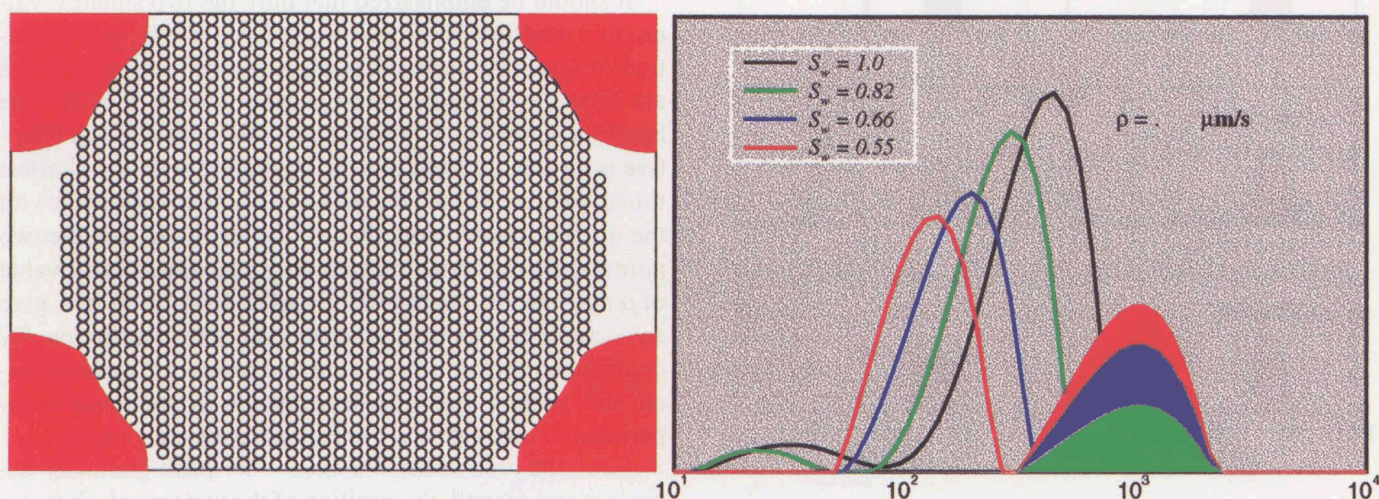


FIG. 7. Distribution of oil blobs (shaded) in the intergranular pores of the model shown in Figure 4. We are looking at a slice through a unit cell centered at the corner of, and oriented at 45° relative to the cell shown in Figure 4. In the right-hand panel the T_2 distributions for four values of S_w are shown. The smallest value of S_w corresponds to the filling of the intergranular pores with oil.

that of bulk oil. It is represented here as a peak centered at $T_2 = 1$ s, which is appropriate for a light oil. For $S_w = 0.55$ the entire macropore is filled with oil and diffusion between the micro- and macropores is completely suppressed. The T_2 distribution then separates into two well-defined peaks corresponding to oil and water. For $0.55 \leq S_w \leq 0.66$ the shape of the oil blob is as shown in Figure 7. For $S_w \geq 0.66$, the area under the oil peak decreases and the part of the distribution associated with water molecules regains the structure associated with coupling between inter- and intragranular pores.

ANALYTICAL MODELS

General Equations

In Figure 8 we depict, schematically, the structure of two simpler models that capture the physical picture developed in the preceding section. Here we envision either a 3D array of spherical microporous grains each of which is surrounded by intergranular pore space or a 1D periodic arrangement of identical slab-like microporous grains separated by intergranular pores.

The central equations that describe these two models are identical. Within the intergranular pore, the magnetization, M_m , is subject to the usual diffusion equation:

$$D_0 \nabla^2 M_m - \frac{M_m}{T_{2B}} = \frac{\partial M_m}{\partial t}. \quad (5)$$

The microporous grains are treated as a continuum with local magnetization per unit volume of pore space, M_μ . In contrast to the model of Figure 4, no attempt is made here to represent the actual structure of the microporosity. Instead, this structure enters through the local porosity, ϕ_μ and the electrical formation factor, F_μ , of the microporous

continuum. Enhanced relaxation at the interface within the microporous grain is represented in terms of the decay rate $1/T_{2\mu} = \rho/V_{S\mu}$, where $V_{S\mu}$ is the pore-volume-to-surface-area ratio within the microporous grain. Within the microporous grain, M_μ obeys the modified diffusion equation

$$\frac{D_0}{F_\mu} \nabla^2 M_\mu - \phi_\mu M_\mu \left(\frac{1}{T_{2\mu}} + \frac{1}{T_{2B}} \right) = \phi_\mu \frac{\partial M_\mu}{\partial t}. \quad (6)$$

What are the boundary conditions to be applied at the interface between the macro- and micropores? First, there is the simple continuity condition, $M_m = M_\mu$. We must also match the flux from the macropore into the wall with (1) the flux away from the wall in the microporous grain and (2) a surface demagnetization rate dependent on a local surface relaxivity ρ :

$$D_0 \mathbf{n} \cdot \nabla M_m = \frac{D_0}{F_\mu} \mathbf{n} \cdot \nabla M_\mu + \rho M_m (1 - \phi_\mu) \quad (7)$$

where $\mathbf{n}$ is the unit normal vector pointing out of the grain. Note that if we let $\phi_\mu \rightarrow 0$, equation (7) reduces to $D_0 \mathbf{n} \cdot \nabla M_m = \rho M_m$ and we recover the problem studied by Brownstein and Tarr (1979). The final boundary condition is the periodicity of M_m across the faces of the unit cell.

Equations (5) through (7) determine the behavior of the two models defined in Figure 8. The solution to these equations can be obtained in a variety of ways, for example, by an extension of standard eigenfunction methods or by Laplace transforms. The computational advantage of one method over the other depends on the specific details of the problem. Both models are defined by the same four geometrical parameters introduced in connection with Figure 4. For example, in the 1D case, the intergranular porosity is given by $f_m = L/W$ and the average microporosity of the system is $f_\mu = \phi_\mu f_g = \phi_\mu (1 - f_m)$. The pore volume to surface area ratio for the intergranular pore, V_{Sm} , is simply L while that for the microporous grain is $\rho T_{2\mu}$. While the 3D model might appear to have an additional curvature parameter, R_g , this model is in fact fully defined by ϕ , f_m , V_{Sm} , and $T_{2\mu}$. For both models, the physical parameters D_0 , T_{2B} , and F_μ must be estimated independently.

Apparent Relaxivity

The boundary condition of equation (7) allows us to introduce the concept of *apparent* relaxivity, ρ_a at the interface between the intergranular pore and the microporous grain. Suppose we assume that the magnetization near the surface of the microporous grain varies on a time scale much larger than $T_{2\mu}$. The transient term in equation (6) can then be ignored to yield a *quasistatic* description of M_μ , and ρ_a can be defined by writing the boundary condition (7) as

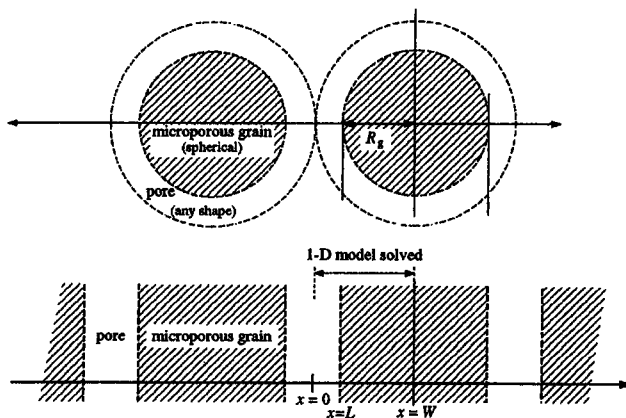


FIG. 8. Three-dimensional (upper panel) and one-dimensional (lower panel) geometries underlying our analytical models.

$$D_0 \mathbf{n} \cdot \nabla M_m = \frac{D_0}{F_\mu} \mathbf{n} \cdot \nabla M_\mu + \rho M_m (1 - \phi_\mu) \equiv \rho_a M_m. \quad (8)$$

Solving the quasistatic equations, the formulas for ρ_a in one and three dimensions are:

$$\rho_a^{(1)} \approx \rho(1 - \phi_\mu) + \sqrt{\frac{\phi_\mu D_0}{F_\mu T_{2\mu}}} \tanh \left(\sqrt{\frac{\phi_\mu F_\mu L_g^2}{D_0 T_{2\mu}}} \right), \quad (9)$$

where $L_g \equiv W - L$ represents a one-dimensional “grain radius,” and

$$\rho_a^{(3)} \approx \rho(1 - \phi_\mu) + \sqrt{\frac{\phi_\mu D_0}{F_\mu T_{2\mu}}} \coth \left(\sqrt{\frac{\phi_\mu F_\mu R_g^2}{D_0 T_{2\mu}}} \right) - \frac{D_0}{F_\mu R_g}. \quad (10)$$

Substituting typical numbers for reservoir grainstones and packstones ($D_0 = 2.5 (\mu\text{m})^2/\text{ms}$, $\phi_\mu = 0.20$, $F_\mu = 25.0$, $R_g = 100 \mu\text{m}$, and $\rho = 1.5 \mu\text{m/s}$), we find (1) that the diffusion length within the microporous grain,

$$\sqrt{D_0 T_{2\mu} / (\phi_\mu F_\mu)},$$

is much smaller than the grain diameter (so that the penetration of M_m into the microporous grain is essentially a surface effect) and (2) that equations (9) and (10) reduce to

$$\rho_a \approx \sqrt{\frac{\phi_\mu D_0}{F_\mu T_{2\mu}}} \quad (11)$$

Note that the microscopic parameter, ρ , does not appear explicitly in equation (11); it enters only through the relaxation time, $T_{2\mu}$, a quantity that can, in principle, be measured directly. With the numbers given above, we find that the value of ρ_a can easily be an order of magnitude larger than the intrinsic ρ value.

The above equation for ρ_a immediately suggests an approximate two-exponential fit to the relaxation data

$$M(t) \approx (\phi - f_m) \exp \left(-\frac{t}{T_{2\mu}} \right) + f_m \exp \left(-\frac{\rho_a t}{V_{Sm}} \right). \quad (12)$$

Here V_{Sm} is the ratio of the macropore volume to the external surface area of the microporous grain. We emphasize that equation (12) suffers from two important drawbacks. First, because ρ_a is generally much larger than ρ , the time scale for macropore relaxation is not significantly different from $T_{2\mu}$ and the quasistatic approximation is not justified. Second, when the diffusion length in the microporous grain is not small, a significant amount of magnetization can leak from the microporous grain to the intergranular pore. The amplitudes of the dominant decay modes are then *not* sim-

ply related to the porosity fractions as in equation (12). In practice, we find that the two-exponential fit overestimates the value of f_m . For the 1D model, an explicit, but approximate, correction is given in equation (16).

On the other hand, for intermediate values of ρ , when

$$\sqrt{D_0 T_{2\mu} / (\phi_\mu F_\mu)} \ll R_g,$$

the two-exponential approximation is more realistic. As ρ decreases, however, some of the microporosity leaks into the intergranular pore and this picture breaks down. Mathematically, this breakdown arises because the two-exponential approximation does not satisfy the boundary condition (8). Thus, while the quasistatic approximation explains the enhanced relaxivity of the intergranular pores, it is inadequate for the ρ values of interest in reservoir carbonates.

One-dimensional analytical results

Without any additional approximations, the equations developed above can be solved by introducing a Laplace transform of $M(t)$ to $\bar{M}(s)$

$$\bar{M}(s) \equiv \int_0^\infty M(t) e^{-st} dt. \quad (13)$$

Once $\bar{M}(s)$ has been evaluated, inversion back to $M(t)$ can be carried out either numerically or using contour integration. The time-dependent magnetization is evaluated from the residue theorem:

$$M(t) = \sum_{n=0}^{\infty} \lim_{s \rightarrow s_n} (s - s_n) \bar{M}(s_n) e^{s_n t}, \quad (14)$$

where (s_n) are the simple poles of $\bar{M}(s)$ in the complex s -plane. $M(t)$ is thus a sum of discrete exponential decays with amplitudes that may be evaluated from equation (14). To each mode rate there corresponds a *shape*; these can be used to compute explicit magnetization profiles through the pore and grain, at various stages of the decay.

Typical results obtained for this model are illustrated in Figure 9. Here we have taken $\rho = 1.50 \mu\text{m/s}$, and fixed the geometrical parameters to match the system described in Figure 6. In Figure 9a we see that the $M(t)$ curves calculated from the one-dimensional model and the three-dimensional random walk simulations are essentially identical. In Figure 9b, for five different times, ($t = 0, T_{2\mu}/2, T_{2\mu}, T_0$, and $2T_0$) the magnetization is shown as function of position in the one-dimensional unit half cell. (T_0 is defined in equation (15).) Here we see that the intergranular pore exhibits fast diffusion behavior (i.e., spatially uniform magnetization) while the microporous grain shows a clear magnetization gradient. Physically, this gradient represents the replenishing of magnetization within the grain via

the diffusive entry of water molecules coming from the intergranular pore. In Figure 9c the direct output of the normal mode solution is illustrated. The uncoupled case, determined simply by the input geometry, is represented by

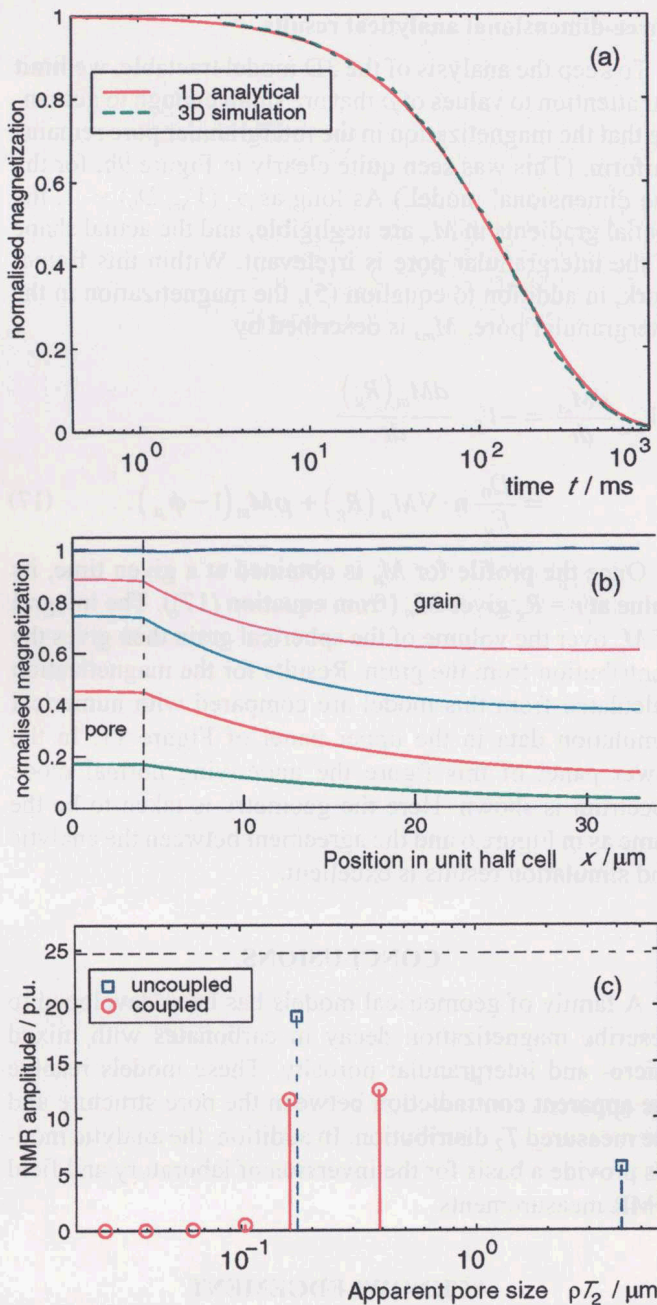


FIG. 9. Results for the one-dimensional analytical model with parameters chosen to correspond to those of Figure 6. In panel (a) the magnetization, $M(t)$, is compared with the results obtained from the random walk simulation. In (b) the spatial dependence of the magnetization is shown for five values of t (see text). $W = 73.5 \mu\text{m}$, so that only a fraction of the grain is shown.) In panel (c) the location of the normal modes is displayed with the uncoupled and coupled modes shown in blue and red, respectively.

two sticks ending with boxes, whose height gives the amplitude of the corresponding mode. The coupled case is dramatically different. Here the modes are shown as sticks ending with circles and the enhanced relaxivity of the microporous grain acting on the magnetization in the intergranular pore has shifted the corresponding mode to much earlier times. In addition, we see the *spillover* effect (i.e., penetration of magnetization from the intergranular pore into the grain) which markedly increases the amplitude of the pore mode. In Figure 10 the one dimensional normal mode results are compared directly with the T_2 distributions from Figure 6. The level of qualitative agreement is quite satisfying.

Some simple approximations help in understanding these results. In many typical cases the decay is dominated by just two or three exponential decay modes. There is usually a single *pore-like* mode, which arises mainly from the intergranular pore, and which penetrates exponentially into the grain. Its relaxation rate, $1/T_0$, is given approximately by

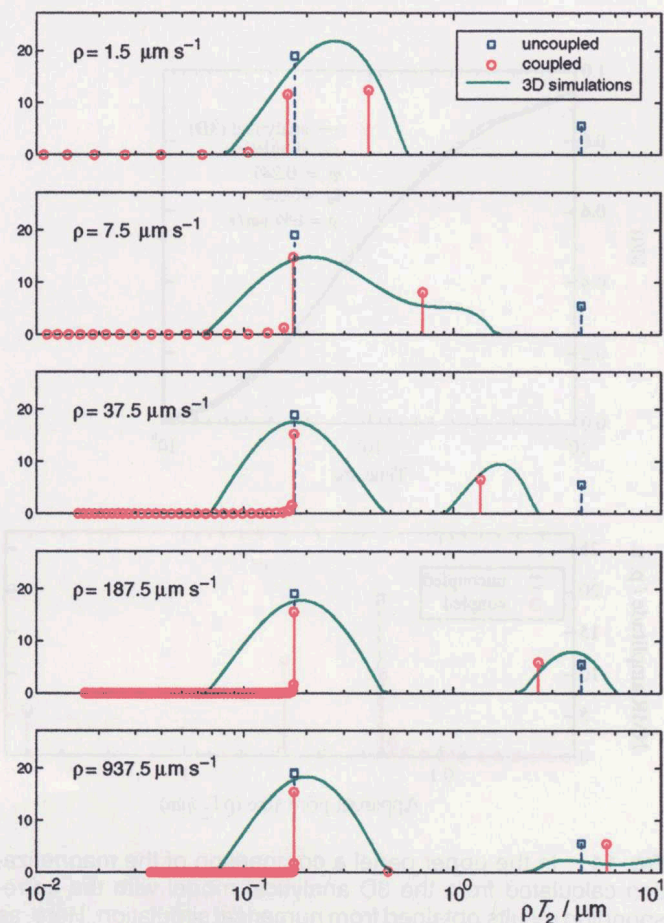


FIG. 10. Modal analysis from the 1D coupled diffusion model (red) compared with the T_2 distributions shown in Figure 6 (green).

$$\frac{1}{T_0} \approx \frac{1}{L} \left[\sqrt{\frac{\phi_\mu D_0}{T_{2\mu} F_\mu}} + \rho(1 - \phi_\mu) \right] \equiv \frac{\rho_a}{L} \quad (15)$$

Here we see that the rate assigned to this mode is controlled by the apparent relaxivity, ρ_a , originally defined in connection with equation (9). Its intensity I_0 is shifted from f_m to

$$I_0 \approx f_m \frac{(1 + X)^2}{1 + X/2} \quad (16)$$

where

$$X \approx (T_{2\mu} / T_0) / \sqrt{1 - T_{2\mu} / T_0}.$$

Because of the strength of the diffusive coupling in shifting T_0 from the intrinsic value, L/ρ , towards $T_{2\mu}$, X is not necessarily small. The rest of the decay is often dominated by one or two "grain-like" modes, shifted by small amounts away from $T_{2\mu}$. The practical consequence is that frequently the magnetization can be approximated as a bi- or tri-exponential decay. However, the rates and intensities of such modes are related in a complicated and nonlinear way

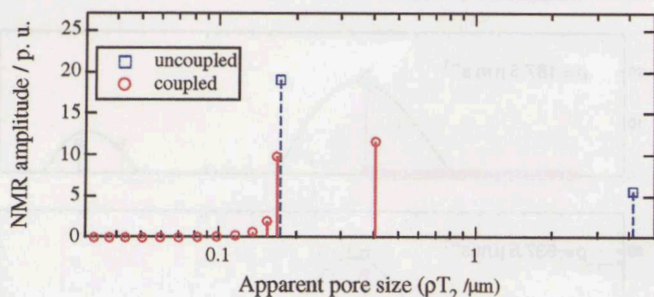
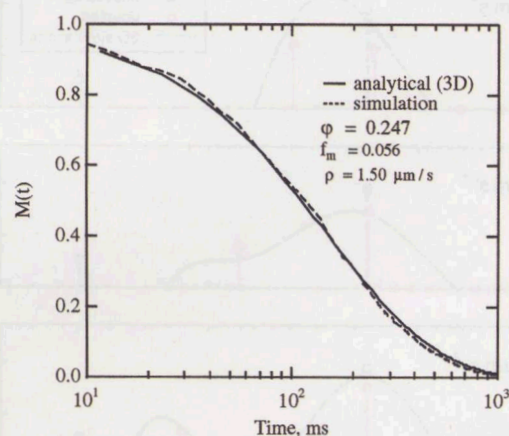


FIG. 11. In the upper panel a comparison of the magnetization calculated from the 3D analytical model with the corresponding results obtained from numerical simulation. Here, as in Figure 9, the geometrical parameters are chosen as in Figure 6. In the lower panel the corresponding normal modes are shown. These results are very similar to their one-dimensional counterparts shown in Figure 9c.

to the main parameters of the model, such as the porosity partitions f_m and ϕ_μ , and the length scales L and $\rho T_{2\mu}$. We emphasize again that approximations like equation (12) are generally not valid for the analysis of carbonate NMR data.

Three-dimensional analytical results

To keep the analysis of the 3D model tractable, we limit our attention to values of ρ that are small enough to guarantee that the magnetization in the intergranular pore remains uniform. (This was seen quite clearly in Figure 9b, for the one dimensional model.) As long as $\rho_d/(V_{sm} D_0) \ll 1$, the spatial gradients in M_m are negligible, and the actual shape of the intergranular pore is irrelevant. Within this framework, in addition to equation (5), the magnetization in the intergranular pore, M_m , is described by

$$\begin{aligned} -V_{sm} \frac{dM_m}{dt} &= -V_{sm} \frac{dM_m(R_g)}{dt} \\ &= \frac{D_0}{F_\mu} \mathbf{n} \cdot \nabla M_\mu(R_g) + \rho M_m(1 - \phi_\mu). \end{aligned} \quad (17)$$

Once the profile for M_μ is obtained at a given time, its value at $r = R_g$ gives M_m (from equation (17)). The integral of M_μ over the volume of the spherical grain then gives the contribution from the grain. Results for the magnetization calculated from this model are compared with numerical simulation data in the upper panel of Figure 11. In the lower panel of this figure the underlying normal mode spectrum is shown. Here the geometry is taken to be the same as in Figure 6 and the agreement between the analytic and simulation results is excellent.

CONCLUSIONS

A family of geometrical models has been developed to describe magnetization decay in carbonates with mixed micro- and intergranular porosity. These models resolve the apparent contradiction between the pore structure and the measured T_2 distribution. In addition, the analytic models provide a basis for the inversion of laboratory and field NMR measurements.

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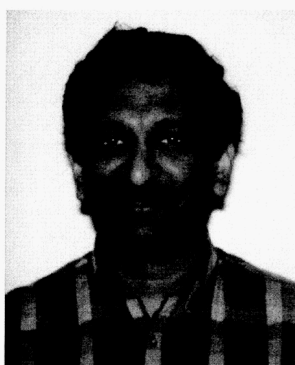
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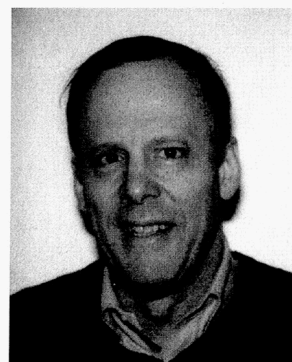
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